

NOTES FROM

How Bad Do You Want It?: Mastering the Psychology of Mind over Muscle

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Central Governor Model (CGM). This model proposes that endurance performance is limited by a subconscious intelligent system in the brain (the central governor) that regulates locomotor muscle recruitment so that the speed/power output sustained over a race never exceeds the capacity of the body to cope with the stress of endurance exercise

This ideal goal would also need to be sufficiently well defined to pull the athlete beyond past limits, yet somehow vague enough that it did not place an artificial ceiling on the athlete's performance.

Bracing yourself—always expecting your next race to be your hardest yet—is a much more mature and effective way to prepare mentally for competition.

technique of reminding herself how much she preferred the suffering of running hard all the way to the finish line to the shame of giving up could not have worked for her if she hadn't lived through the experience of choking on many occasions and become sick and tired of it

Caffeine, music, and transcranial electrical stimulation do not conserve glycogen or neutralize lactic acid or enhance physical capacity in any other way. Yet they do enhance endurance performance. Acting on the brain instead of the muscles, these things work by making exercise feel easier

Second, in the new endurance sports psychology, the role of sports psychologist is taken on by you—the athlete. Why? Because the only way to become really good at coping with the discomforts and stresses of endurance sports is to experience them. Visualization sessions and goal-setting exercises alone won't help your mind quell your body's rebellion in the toughest moments of a race. Developing mental fitness requires exposure to these challenges no less than the development of physical fitness requires exposure to workouts. Nobody can do this work for you, or even guide you through it. Coping is a response to discomfort and stress, after all.

It turns out the essential challenge of endurance sports really is psychological.

perception of effort and other psychological factors affect endurance performance

self-consciousness reduces endurance performance, and we know from other research that perceived pressure promotes self-consciousness

By keeping track of, and aiming to improve, personal best times for specific race distances, athletes can exploit this phenomenon to try harder than they would otherwise be able to

In common language, this attitude of acceptance toward an impending disagreeable experience is called “bracing yourself.” Many of us use this coping skill instinctively to reduce the unpleasantness of everyday trials such as a trip to the dentist’s office

an athlete cannot truly maximize her mental fitness without going beyond the acquisition of context-specific coping skills and improving her overall coping style—by changing as a person—to some degree.

Hard physical limits do exist, of course, but no athlete ever reaches them because the purely psychological limit of perceived effort tolerance is always encountered first

Other factors increase the amount of perceived effort an athlete can (or will) tolerate. An obvious example of this sort is motivation

In mastering my fear of suffering in races, I acquired a greater level of respect for myself, a sense of inner strength that has helped me tackle other challenges, both inside and outside sports.

Neuroplasticity is a hardwired coping skill, operating automatically whenever an athlete practices an old skill under a new constraint. But in order to benefit from the workaround effect, an athlete must first exercise another, psychological coping skill, one that requires a conscious choice and that Willie Stewart relied on intuitively: adaptability.

self-consciousness distracts athletes from the task of the moment such that, as Beilock puts it in *Choke*, they “are not devoting enough attention to what they are doing and rely on simple or incorrect routines.”

Even though all of the students did the same training, those who pursued quantitative goals improved their performance significantly more when the muscular endurance test was repeated after eight weeks.

acceptance reduces the unpleasantness of pain without reducing the pain itself. For this reason, it is a more effective coping skill. The same skill also reduces perceived effort

physical fitness comes only partly from training—the rest comes from natural inheritance

Sammy had overcome physical weakness with psychological fortitude

Endurance athletes, by definition, endure. They endure long hours of training, the privations of a monastic lifestyle, and all manner of aches and pains. But what endurance athletes must endure above all is not actual effort, but perception of effort.

I'd been moonlighting as a miler for the Oyster River Middle School track team and having some success with it at the time of my injury. So I decided to put all my chips on running.

Physical preparedness is one such factor. Research has demonstrated that people are more likely to experience flow when performing tasks in which they have a high level of mastery. Other investigations have revealed that well-trained athletes exhibit reduced activity in parts of the brain that are related to physical self-consciousness. These things are linked. Well-trained athletes have an easier time achieving flow because they are less physically self-conscious.

college students shot basketball free throws more accurately when they were asked to concentrate on the back of the rim than they did when they were told to focus on snapping the wrist when the ball left their hand (an element of “good technique”). Psychological pressure has a similar effect, inducing a state of self-consciousness that effectively shifts the athlete’s attention from the “rim” to the “wrist.” Endurance sports do not require the same kind of coordination that games like basketball do, but self-consciousness is known to reduce movement efficiency in running, swimming, and other forward-motion activities.

humans cannot sustain maximum-intensity exercise longer than about 30 seconds without exceeding the highest level of perceived effort they can tolerate

Whether an athlete is able to find a solution to such a challenge and become mentally fitter in the process depends on his overall mindset toward the sport. Athletes who consciously intend to use their experiences to develop their mental fitness tend to acquire better coping skills more quickly than do athletes for whom becoming a better fire walker is not an explicit goal

When cycling or running or rowing at race intensity feels easier, athletes are able to push closer to their true physical limit before they max out their tolerance for suffering

encountering them with prior knowledge of their nature and of the methods of coping with them that have proven most effective for other athletes

except in extreme cases, dehydration, which is biological, does not cause athletes to slow down in races; instead, the psychological condition of feeling thirsty does

becoming your own “sport psychologist.”

Siri Lindley did not have PTSD, but she had an overactive internal critic that was probably related to the trauma of her parents’ divorce and was almost certainly the source of her tendency to choke under pressure

This is why as a coach I have to try to understand the background that an athlete is coming from on a mental perspective/level.

It’s not the time goal itself that enhances performance but the effect that the goal has on how the athlete interprets her perception of effort.

people often choose to expect the worst of an upcoming experience in hopes of creating a more favorable contrast between their expectations and reality.

Psychologists make a distinction between skills and traits. A trait is essentially a generalized, or non-context-specific, coping skill. Examples of coping traits are general self-efficacy, or belief in one's overall competence, and internal locus of control, or belief that one is the captain of one's ship rather than a mere puppet of fate. No athlete can maximize her mental fitness without enhancing traits like general self-efficacy and internal locus of control in addition to gaining context-specific skills such as a refined sense of pace.

perception of effort is an athlete's sense of how hard she is working. It is distinct from pain, fatigue, proprioception, and other perceptions that athletes experience when racing, and it is the primary source of the discomfort that causes athletes to slow down or quit when they hit the wall in races.

Psychologists use the term coping to refer to a person's behavioral, emotional, and cognitive responses to discomfort and stress. Endurance sports are largely about discomfort and stress; hence they are largely about coping. In a race, the job of the muscles is to perform. The job of the mind is to cope. But here's the hitch: The muscles can only perform to the degree that the mind is able to cope. Endurance sports are therefore a game of "mind over muscle."

Talent alone doesn't cut it. This realization filled me with a mix of healthy shame and inspiration that moved me to try harder.

the organ that limits endurance performance is the brain, not the cardiovascular system and fatigued locomotor muscles

adaptability begins with the belief that practicing an old skill is worthwhile despite the existence of a new constraint

Athletes move more efficiently when their attention is focused on key features of their environment rather than on their own body

those runners who end up closest to these round numbers at the finish line slow down less than other runners in the final miles of a marathon—evidence that the pursuit of these round-number goals enhances performance

Except in the case of reflex actions, all muscle work begins with an act of conscious willing

Mental fitness is also partly innate. Some athletes are natural fire walkers.

The physiological factors that limit performance in a marathon are well known. One of these factors is depletion of glycogen fuel stores in the working muscles

One cannot improve as an endurance athlete except by changing one's relationship with perception of effort.

A part of me seemed to ask, How bad do you really want this?, and another part answered, Not as much as that guy

Perhaps not surprisingly, what sets Siri apart from other coaches is her heavy emphasis on the mental side of the sport. Having worked through the psychological issues that held her back as an athlete, Siri believes she has a formula to help other athletes do the same. A single word is tattooed on each of her wrists: “GRATITUDE” on the left wrist and “BELIEVE” on the right. These two words encapsulate Siri’s coaching philosophy.

IF CHOKING IS A condition of heightened self-consciousness that intensifies perceived effort and hampers endurance performance, then its opposite would be a mental state in which self-consciousness disappears, reducing perceived effort and boosting performance. Does such a thing exist? It does. The psychologist Mihály Csikszentmihalyi called it “flow” and defined it as a state of complete immersion in a purposeful activity. Endurance athletes describe the flow state as one in which they seem to become the thing they are doing. The part of the mind that normally watches the part that is focused on the task at hand vanishes, leaving the athlete’s consciousness directed externally in a way that feels right and yields exceptional performance

But when an athlete starts a race that he knows will last longer than 30 seconds, he holds back just enough that his perceived effort limit is not reached until he is at the finish line. That is the art of pacing.

Those who were most motivated by personal goal achievement, self-esteem, or finding meaning in life were far less likely to quit than were those whose primary goal was weight loss or social recognition

No matter what you do, you will always have reserve physical capacity at the end of any race or time trial that lasts longer than 30 seconds, give or take

equip athletes with this knowledge so that they do not have to “reinvent the wheel” in their attempts to master the discomfort and stress of their sport but can function successfully as their own sports psychologist.

Mind is everything. Muscle—pieces of rubber. All that I am, I am because of my mind

the power of psychology has not been fully exploited in endurance sports, and to use it deliberately and systematically is one of the main practical suggestions derived from the Psychobiological Model

silence her internal critic

One of the goals of meditation is meeting, understanding, and silencing your internal critic

Setting time-based goals that stretch you just beyond past limits is like setting a flag next to a bed of hot coals to mark the furthest point reached in your best fire walk. That flag says to you, “This is possible, and you know it. So why wouldn’t it be possible for you to make it just one step farther the next time?”

The great British runner Mo Farah told a reporter for The Daily Mirror ahead of his first marathon, “This will be the hardest race of my life.” He wasn’t being negative; he was bracing himself.

endurance racing as “a test of you as a person on top of a test of you as an athlete.”

The neurophysiology of perception of effort is complex and not yet fully understood, but it appears to be closely linked to the intensity of activity in parts of the brain that drive the muscles to contract

successful coping in endurance sports is any response to discomfort and stress that favorably changes an athlete’s relationship to perception of effort, either by increasing the amount of effort the athlete is able to give or by enhancing what the athlete gets out of her best effort.

Hunter recognized that he wanted to win the race more than he feared his legendary challenger, and enough to suffer for it.

there is no evidence to indicate that a central governor exists inside our brains, and endurance athletes have considerable control over their performance. This alternative model of how the brain regulates endurance performance is called the Psychobiological Model. Its core principles are that decisions about pacing or quitting during endurance competitions are taken by the conscious brain and that these decisions are primarily based on the conscious sensation of how hard, heavy, and strenuous exercise is, a feeling we call perception of effort.

Adaptability is responding with a similar mindset whenever a physical setback occurs. If you do, your body and mind will follow suit, the body adapting by finding new efficiencies and the mind by elevating mental fitness through heightened focus or inhibitory control

Simply put, self-consciousness increases perceived effort. Racing in a state of pressure-induced self-consciousness is like fire walking while giving your full attention to the painful heat in your feet rather than focusing on where you're going. As an athlete, you're much better off directing your attention externally, to the task at hand, which distracts you to some degree from your suffering, allowing you to push a little harder.

Regardless of how an athlete chooses to train, her training will yield greater improvement in race times if improving race times is the explicit goal of the training process.

movement-related cortical potential (MRCP). Marcora has shown that MRCP and perception of effort are high when subjects exercise at maximum intensity and also that they increase covariantly when exercise of lower intensity is performed for a long period of time.

endurance performance is a species of self-regulation, or the process by which organisms control their internal states and behavior in the pursuit of their goals

Until fairly recently, exercise scientists would have said that a physically weaker athlete could not defeat a stronger rival by virtue of greater mental fitness

The fitter the athlete becomes, the easier it feels for her to swim, bike, run, or whatever at any given speed, and that is why her performance improves

I loved running, loved getting fitter and faster, but I hated suffering the way I did in races.

“Gratitude” is about letting go of desired outcomes and fully embracing the privilege and process of pursuing goals and dreams. “Believe” refers to the confidence that arises naturally through this process, a self-trust that is the antithesis of the doubt-fueled fixation on goals and dreams expressed in Siri’s nightly fantasy of having the perfect race at the 2000 Olympics.

self-belief has to come first. So where does it come from? It comes from letting go, as illustrated by Siri’s surrender to Brett Sutton’s methodology. Counterintuitive though it may be, caring a little less about the result of a race produces better results. An athlete who believes in herself whether she succeeds or fails is able to put her goal out of mind and race in the moment, and to race in the moment—in flow—is to race better. The athlete who lacks self-belief can gain it by consciously pushing her goals and the worries that surround them out of her mind and teaching herself to stay focused on the task of the moment throughout the training process that leads up to the next big race.

One of the most important and valuable coping skills in endurance sports is the ability to interpret the perceptions that influence pacing decisions in a performance-maximizing way.

The overarching intention that all competitive endurance athletes share is improved performance. But maximizing one’s mental fitness is instrumental to this goal. Therefore the

intention of maximizing mental fitness should be as consciously held as that of getting faster.

An athlete who gets just a bit closer to his physical limit in comparison to another athlete who is somewhat stronger may indeed defeat the stronger athlete in head-to-head competition

The methods that the greatest athletes rely on to overcome the toughest and most common mental barriers to better performance are practically by definition the most effective coping methods for all athletes.

From a psychobiological perspective, endurance performance is determined solely by the mind's output; biology is no more than an external input to the mind, influencing its output.

Conscious self-regulation of thoughts, emotions, and behavior can have a dramatic influence on endurance performance, as beautifully illustrated by the real-life examples provided in this book. But genetics, physical training, and nutrition (e.g., carb intake and caffeine) also play a big role because they have profound influences on perception of effort. The subconscious brain can also influence perception of effort, as we recently demonstrated using subliminal visual messages.

“You know what, Siri?” Sutton said. “Forget about all of that. Starting today, you’re retired. The way you look at this sport and the pressure you put on yourself are just all wrong. You started doing triathlon because you loved it. Let’s go back to that. Let’s just see how fit, how fast, and how strong Siri Lindley can be—and have fun doing it.”

There is indeed evidence that the influence of clock watching on endurance performance is two-sided. The same time goal that enhances performance when it is perceived as a target constrains performance when it is perceived as a limit.

The more discomfort an athlete expects, the more she can tolerate, and the more discomfort she can tolerate, the faster she can go

Merely participating in sports does not guarantee that an athlete will develop a more mature and effective coping style

The brain itself becomes fatigued during prolonged exercise, and brain fatigue also increases perceived effort

To become the best athlete you can be, you need to become really good at coping with the characteristic forms of discomfort and stress that the endurance sports experience dishes out, beginning with perceived effort and extending to the many challenges that are secondary to it, such as fear of failure. You must discover, practice, and perfect the coping skills that conquer these challenges most effectively. My own term for a highly developed overall coping capacity in endurance sports is mental fitness.

“psychobiological” model of endurance performance

I consider endurance performance to be a self-regulated behavior on which thoughts and feelings can have profound influences

They reported being most internally focused when the screen was blank, offering no visual stimulation.

External visual stimulation leads to decreased perceived effort (this is why treadmill running for longer distances can be so tough)

This is why from a coaching perspective, I plan on focusing on form and the small things (weaknesses) early on, and then transitioning to focusing on strengths closer to the race. This way, the more self-conscious focus during training occurs in the earlier portion of the cycle and the athlete is able to imitate these actions subconsciously in the weeks leading up to a race, and ultimately during the race.

If the goal seems to fall out of reach at any point during the race, the athlete is likely to back off her effort. If the goal seems attainable, but only with increased effort, the athlete is likely to increase her effort, provided she's not already at her limit

Perceived effort actually has two layers. The first layer is how the athlete feels. The second layer is how the athlete feels about how she feels. The first layer is strictly physiological, whereas the second is emotional, or affective. Crudely put, an athlete can have either a good attitude or a bad attitude about any given level of discomfort. If she has a good attitude, she will be less bothered by the feeling and will likely push harder.

The only proof of wanting it more is suffering more

A newer model of endurance performance integrates the body and the mind through the brain. This alternative theory was dubbed the “psychobiological model” by its primary developer, Samuele Marcora

Perceived effort is essentially the body’s resistance to the mind’s will. The fitter an athlete becomes, the less resistance the body puts up. Therefore increased physical capacity is always felt.

I left the racecourse disgusted with myself, knowing I hadn’t left it all out there.

When the body loses the ability to achieve a desired level of performance in the accustomed way, the brain responds by seeking out new ways to get the same level of performance out of the body, ways the athlete might never have discovered otherwise. That’s the workaround effect.

The desire to maximize performance and achieve a particular outcome creates a feeling of pressure. This feeling of pressure compromises performance and ensures that the wanted outcome is not achieved.

As an endurance athlete, you want to get better and better at reading these perceptions in such a way that your internal pacing mechanism functions more and more like an automobile’s range display.

believe that you cannot improve as an endurance athlete except by changing your relationship to perceived effort

Rarely do champion endurance athletes credit their physical capacity for their success. More often, they insist that their advantage lies not in having more to give but rather in being able to give more of what they have

In every race, something within each athlete (something we may now specify as perception of effort) poses a simple question: How bad do you want it? To realize your potential as an athlete, you must respond with some version of this answer: More. And then you have to prove it.

When we say that a person has endured something, what do we mean? We mean that the person has gone through a challenging experience.

Brain Endurance Training, that combines physical training with mentally demanding tasks to stimulate the brain areas involved in self-regulation and to increase resistance to mental fatigue

Perception of effort does not disappear during flow. Hard work still feels hard in this state, but the feeling becomes enjoyable in a way that is difficult to put into words.

students given a “difficult/realistic” goal improved more than those given either an “easy” goal or an “improbable/unattainable” goal

invitation to self-exploration. Not all athletes accept this invitation. If you want to become the best “fire walker” you can be—and more than that, if you want to get the most you can get as a person out of the athletic experience—you will accept it. The journey toward becoming a mentally fit athlete is very much a journey of personal development.

he tired out the brains of subjects with a mental task before they performed an endurance test. The subjects reported higher levels of perceived effort and bonked sooner than they did when they performed the same endurance test without preexisting brain fatigue

The new psychology differs from the old in two key ways. First, it focuses squarely on the development of mental fitness, or of coping skills that directly and indirectly alter an athlete’s relationship to perceived effort in ways that improve performance. It is a psychology of mind over muscle.

new model views mind and body as interconnected, with the body distinctly subordinate

we shouldn’t be surprised that perception of effort (and thoughts related to it) can limit endurance performance. Perceptions are powerful.

an internal attentional focus reduced performance by making the same intensity of pedaling feel harder, while an external focus had the opposite effect.

the amount of effort that an athlete puts into a race is influenced by her perception of the attainability of her goal

Some people have a natural tendency to look ahead to the repetition of a familiar pain stimulus with acceptance. They tell themselves, "This is going to hurt, but no worse than before.

Underdeveloped coping skills cause an athlete to struggle in some way; the experience of struggle provokes an adaptive response from the athlete; this response gives rise, sooner or later, to a more effective coping skill

exhaustion occurs during real-world endurance competition not when the body encounters a hard physical limit such as total glycogen depletion but rather when the athlete experiences the maximum level of perceived effort he is willing or able to tolerate

inhibitory control, or the ability to stay focused on task-relevant stimuli (such as a competitor up ahead) in the presence of distracting stimuli (such as a memory of losing to that same competitor in a previous race)

I achieved something greater—the satisfaction of knowing that for once I had truly left it all out there on the racecourse.

neuroplasticity. The brain is highly plastic: It has almost unlimited ability to reorganize itself in response to roadblocks affecting its normal operations. For example, the brain of someone who loses her sight rewires itself in ways that sharpen the other senses. Something similar happens when an athlete returns to training after suffering an injury with lasting physical effects. Unable to stimulate bodily movement in the way it did previously, the brain explores alternatives, eventually settling on the most effective of these.

the athlete's feeling of being under pressure directs her attention toward internal processes such as body movements and anxious thoughts, and this attentional shift undermines performance in a number of ways.

You want to be as correct as you can possibly be when determining the swiftest pace you can sustain to the finish line without exceeding your perceived effort tolerance.

the only limitations you ever encounter in your sport are mental

Physical fitness determines where the wall that represents your physical limit is placed. Mental fitness determines how close you are able to get to that limit in competition. Mental fitness is a collection of coping skills—behaviors, thoughts, and emotions that help athletes master the discomfort and stress of the athletic experience, mainly by increasing tolerance for perceived effort and by reducing the amount of effort that is perceived at any given intensity of exercise

“Sammy proved his heart today,” he said at a post-race press gathering. In sports, “heart” is a metaphor for mental fitness

But it’s the experience, not the biological effects of cold exposure or sleep deprivation, that must be endured.

I came away from this experience having learned a fundamental truth about endurance sports. While my legs and lungs had put me in a position to win, it was my mind—particularly my ability to absorb the shock of the novel sensations I felt and my willingness to suffer a bit for the sake of winning—that had carried me over the top. The essential challenge of long-distance racing, I understood, was mental.

As this study hints, flow is not entirely controllable and hence cannot be regarded as a coping skill in itself. It appears that a number of factors have to line up just right for flow to occur. Insofar as flow is controllable, anything that helps an athlete race less self-consciously promotes the flow state. And flow, in turn, is the ultimate prophylactic against the self-sabotage of choking. Learning how to exploit these flow facilitators, therefore, is a vital coping skill for all endurance athletes.